



# **aUToronto MI Winter Workshop check-in**

**Autopark**

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# Problem Statement

- Urban parking is time-consuming, inefficient, and stressful
- Drivers spend an average of **107 hours/year** (That's over 4.4 days) in New York City searching for parking.
- Creates an estimated **\$72.7 billion** in annual costs from **wasted** time and fuel in the US alone.

Top 10 cities and U.S. average for annual search time, hours per driver:

|                  |     |
|------------------|-----|
| NEW YORK         | 107 |
| LOS ANGELES      | 85  |
| SAN FRANCISCO    | 83  |
| WASHINGTON, D.C. | 65  |
| SEATTLE          | 58  |
| CHICAGO          | 56  |
| BOSTON           | 53  |
| ATLANTA          | 50  |
| DALLAS           | 48  |
| DETROIT          | 35  |
| U.S. AVG.        | 17  |

Table 1: INRIX Parking Ranking – Hours Spent Searching for Parking

| Rank | U.S. City       | Average 2-Hour Parking Cost<br>(One mile of city center) | On-Street Search Time<br>(mins/trip) | Off-Street Search Time<br>(mins/trip) | Annual Search Time<br>(hours/driver/year) | Annual Search Cost Per Driver | Annual Search Cost Per City |
|------|-----------------|----------------------------------------------------------|--------------------------------------|---------------------------------------|-------------------------------------------|-------------------------------|-----------------------------|
| 1    | New York        | \$33                                                     | 15                                   | 13                                    | 107                                       | \$2,243                       | \$4.3bn                     |
| 2    | Los Angeles     | \$14                                                     | 12                                   | 11                                    | 85                                        | \$1,785                       | \$3.7bn                     |
| 3    | San Francisco   | \$12                                                     | 12                                   | 11                                    | 83                                        | \$1,735                       | \$655m                      |
| 4    | Washington D.C. | \$18                                                     | 10                                   | 9                                     | 65                                        | \$1,367                       | \$329m                      |
| 5    | Seattle         | \$10                                                     | 9                                    | 8                                     | 58                                        | \$1,205                       | \$490m                      |
| 6    | Chicago         | \$22                                                     | 9                                    | 8                                     | 56                                        | \$1,174                       | \$1.3bn                     |
| 7    | Boston          | \$26                                                     | 8                                    | 8                                     | 53                                        | \$1,111                       | \$262m                      |
| 8    | Atlanta         | \$6                                                      | 8                                    | 8                                     | 50                                        | \$1,043                       | \$251m                      |
| 9    | Dallas          | \$6                                                      | 8                                    | 8                                     | 48                                        | \$995                         | \$726m                      |
| 10   | Detroit         | \$9                                                      | 6                                    | 6                                     | 35                                        | \$731                         | \$209m                      |
|      | US              | \$4                                                      | 2                                    | 2                                     | 17                                        | \$345                         | \$72.7bn                    |



# Autopark

Our solution to this problem

Autonomously park the car once the user has left the car.

**Example scenario:** You are rushing to the train station, but have time to look for parking, so go to the entrance, and use Autopark to make the car park itself after you leave.





# How does Autopark look like?

A software in the car's UI and an app

When OEMs use Autopark, their cars will have a new feature in their UI called “Autopark”

- When the user clicks on the feature, they will see all the best parking spots in the area, optimized for cost, area and distance.
- Once the user picks a parking lot, if the lot needs payment, the user will be prompted to pay through their phone (through the Autopark app)
- Once the user leaves the car, the car will autonomously park itself!



# Market Size

- In 2025, market research firms like Straits Research and Precedence Research place smart/autonomous parking value between **\$9.3** and **\$11.2 billion**.
  - <https://straitsresearch.com/report/smart-parking-market>
- Projected to grow to over **\$33 billion by 2033**
  - = **17.4% CAGR** (compound annual growth rate) [17-20% range]

| Research Firm             | 2024 Estimate  | 2025 Estimate               |
|---------------------------|----------------|-----------------------------|
| Straits Research          | \$7.98 Billion | \$9.37 Billion              |
| Precedence Research       | \$9.15 Billion | \$11.18 Billion             |
| The Business Research Co. | \$8.5 Billion  | \$10.27 Billion             |
| IMARC Group               | \$5.7 Billion  | (No 2025, but a lower base) |



# Why is the market growing this much?

- **Urbanization and Congestion:** More people are moving to cities, leading to a massive increase in vehicles, traffic congestion, and parking demand.
- **Environmental Concerns:** Reducing the time spent "cruising" for parking directly reduces fuel consumption and carbon emissions, which is a major goal for cities and corporations.
- **Smart City Initiatives + IoT and Connectivity:** Widespread adoption of IoT sensors, 5G, and cloud computing makes it cheaper and easier to get real-time data on which spots are open





# Stakeholder Analysis

- **Primary Customer (B2B): Automotive OEMs (e.g., General Motors, Ford).**
  - Why: In a feature war, where they need to differentiate their AVs.
  - Our “AutoPark” software module is a high-value, licensable feature they can **build into** their new vehicles.
- **End-User (B2C Experience): The AV Owner / Rider.**
  - Why: Core “customer experience” value. They get a true “curbside-to-curbside” experience. They are dropped off at the door, and the car vanishes → handling the parking task entirely.
- **Key Partners (Data & Infrastructure):**
  - Parking Infrastructure Owners: (e.g., municipalities, shopping malls, airports, commercial garages).
    - Provide the real-time data on which spots are open.
  - Data Aggregators: (e.g., Google Maps).
    - Need their APIs to find available lots, and they need your system to offer a premium “park-for-me” feature.



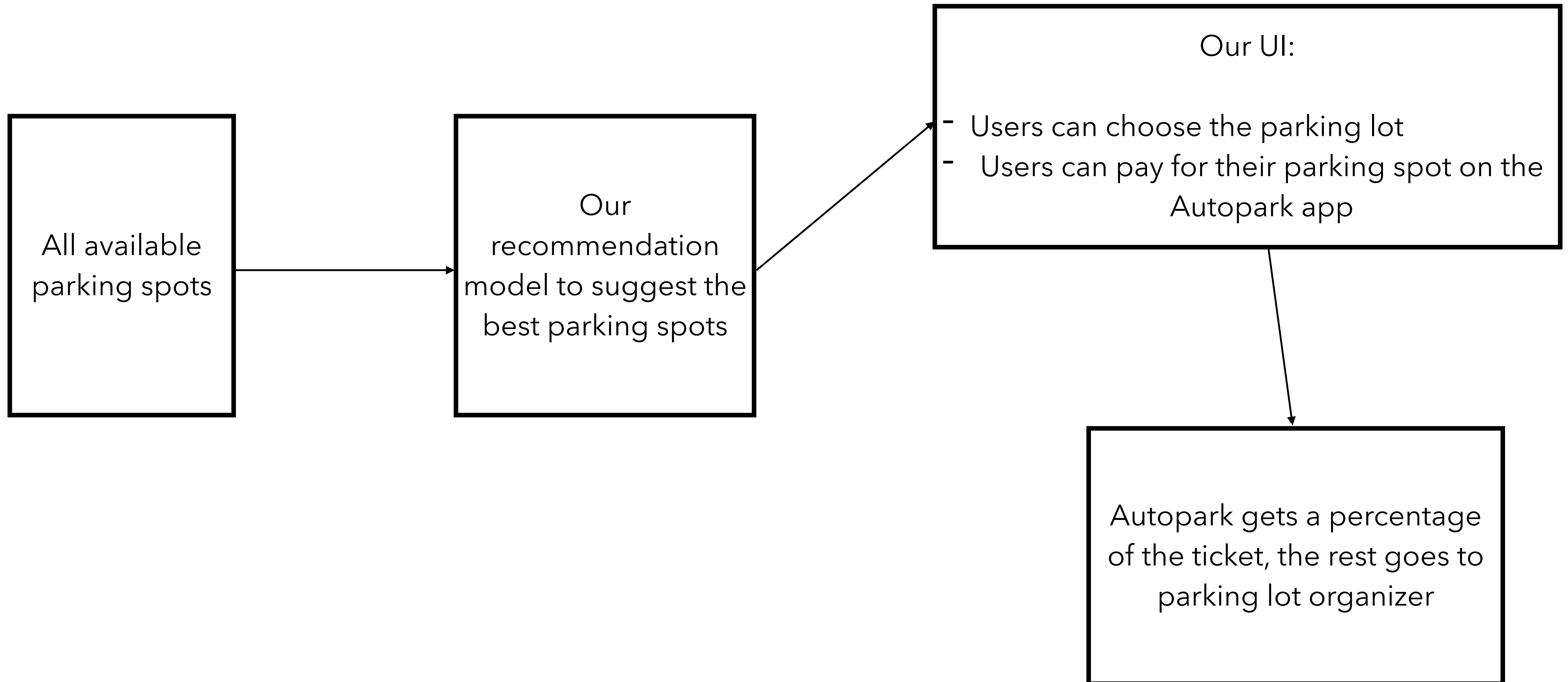
# Revenue Streams

- **SaaS (Software as a Service):** Charge OEMs a small annual subscription fee for V2N network access and map/partner updates, creating recurring revenue.
- **Take a cut of every parking transaction Business-to-Business-to-Consumer (B2B2C):**
  - Up-charge for the parking ticket.
- **The price of the car will be increased with the Autopark feature.**
  - This added cost will be revenue for Autopark.





# High level Overview





# How will parking lots be discovered

Autopark will get parking lot data from Parking Data Aggregators:

- How many spots are available at the parking lot?
- What is cost for a ticket?

Autopark has V2V communication to talk to AVs that are in parking lots.

- Aprox. spots = totals - cars that have been for > 30 mins
- We can also use other indicators like cars moving in circles

This data, along the cost, distance, and location of the parking lots will be fed to our recommendation algorithm.



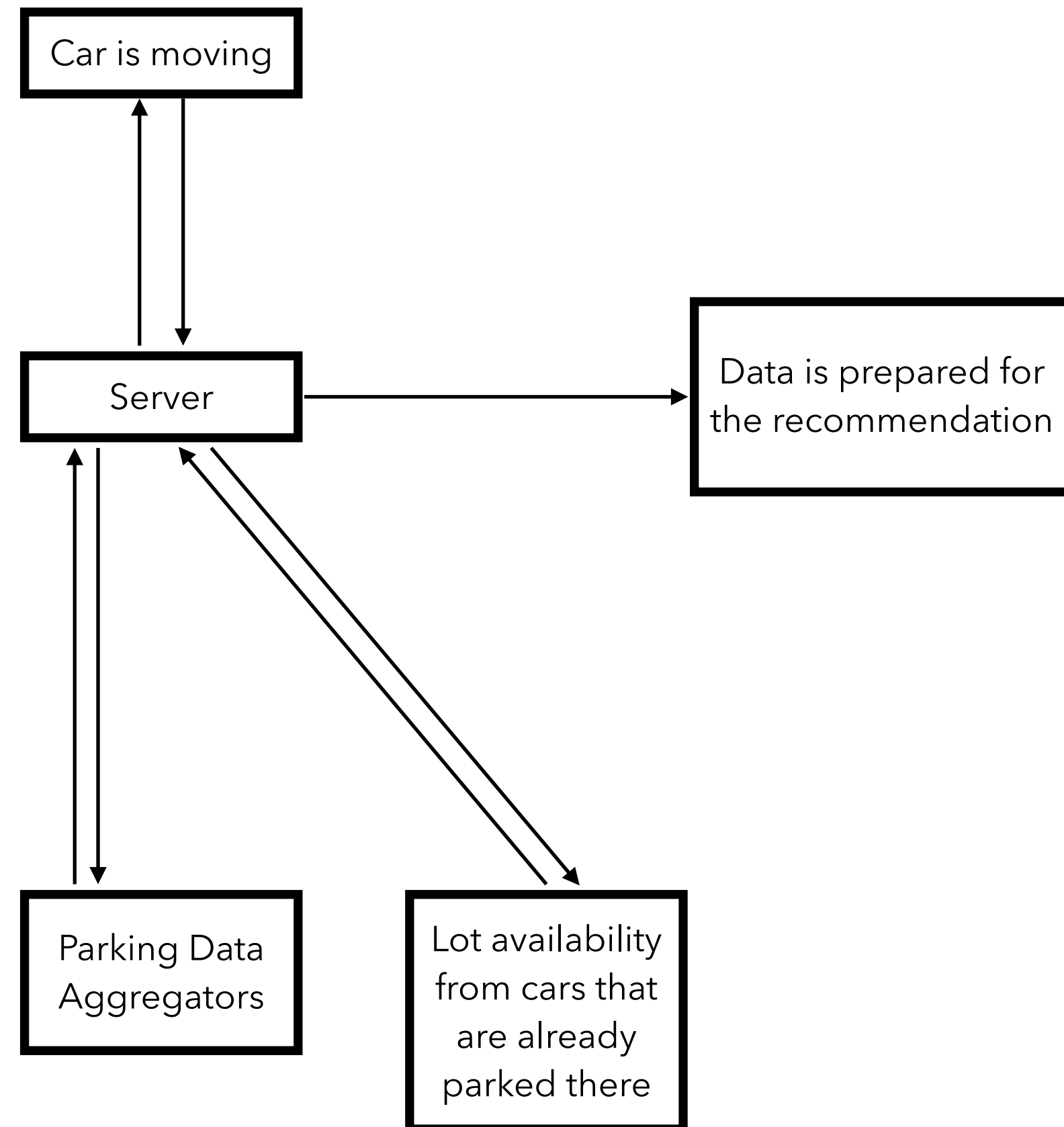
Car is moving

# System Design



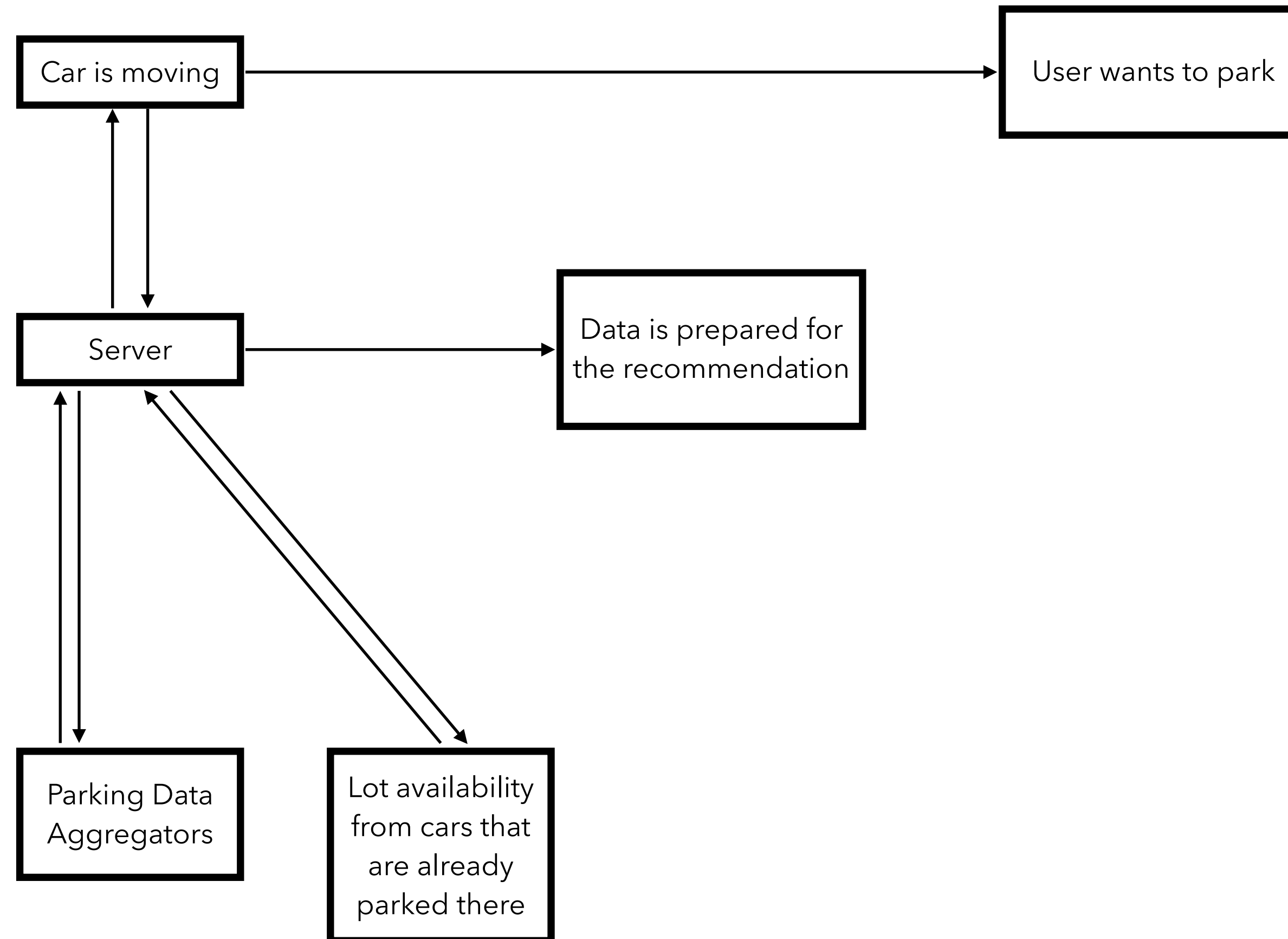


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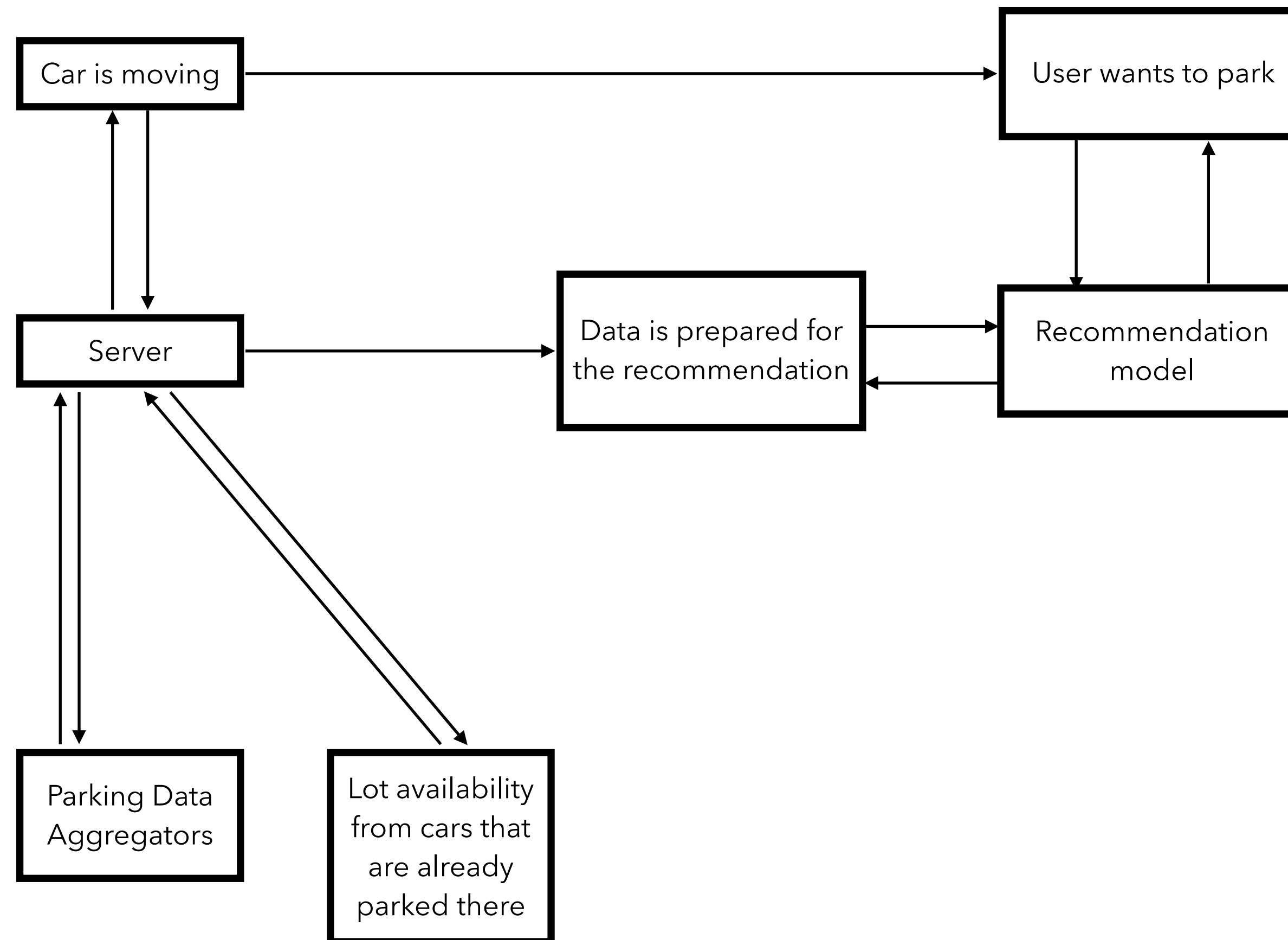


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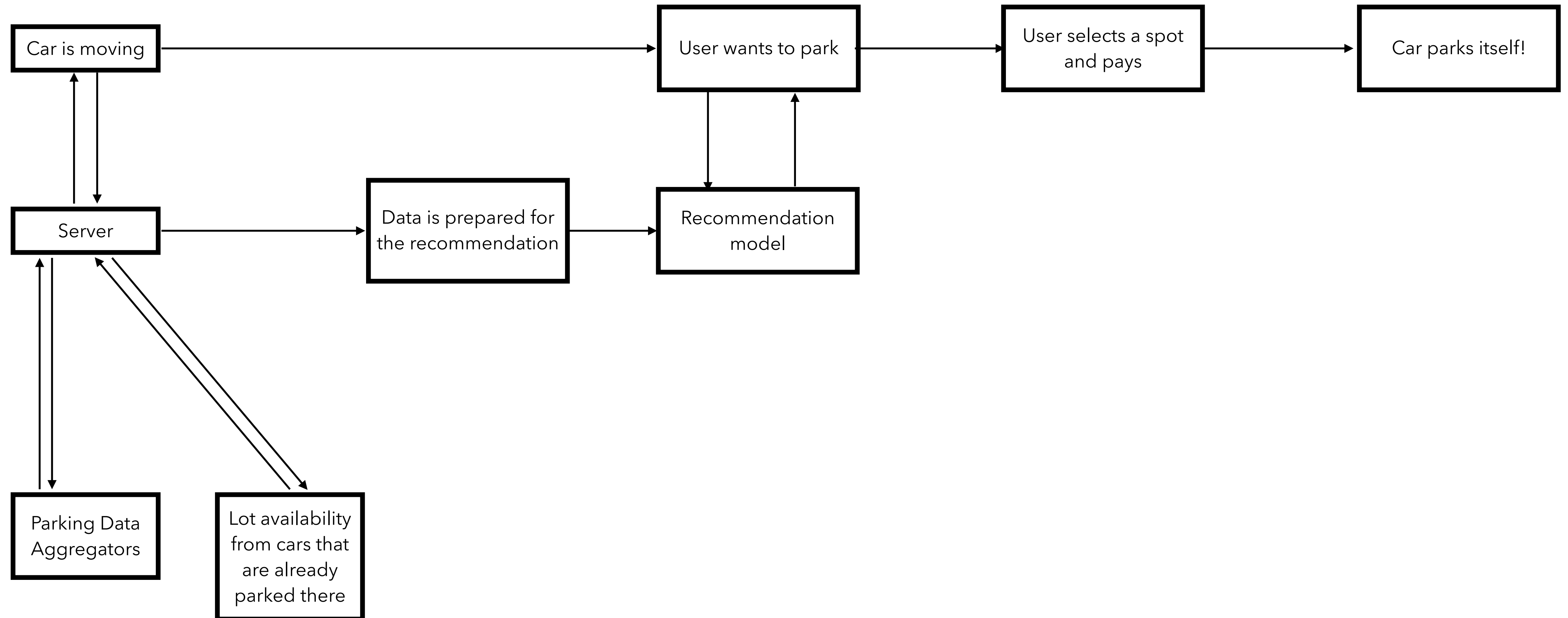
# System Design







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**Any Questions?**